MIL-F-11137C

30 AUGUST 1963

SUPERSEDING
MIL-F-11137B
8 JUNE 1959

MILITARY SPECIFICATION

FILTER, GAS, 150 CFM, M10

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a gas filter for use in removing toxic gases from air forced through the filter.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-P-271 — Paper, Wrapping, Waterproofed Kraft

CCC-T-191 — Textile Test Methods

PPP-B-621 — Boxes; Wood, Nailed and Lock-Corner

PPP-B-636 — Box, Fiberboard

MILITARY

MIL-P-116 — Preservation, Methods of

MIL-A-140 — Adhesive, Water-Resistant, Waterproof Barrier Material

MIL-M-46032 — Mildew - Resistant Treatment (for Fabrics) Copper Processes

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 — Marking for Shipment and Storage

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Varia-

FSC 4240

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bles for Percent Defective

DRAWINGS

EDGEWOOD ARSENAL

DL 5-19-673 — Filter, Gas, 150 CFM, M10 — Data List

(All drawing referenced in this specification are listed on Drawing DL 5-19-673.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All material cited on Drawing DL 5-19-673 or on the subsidiary drawing shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

3.1.1.1 *Cotton cloth.*

3.1.1.1.1 *Mildew resistance.* The cotton cloth shall be made mildew resistant in accordance with Specification MIL-M-46032, using type III, method optional process except that leaching shall be omitted. No visible growth of mildew shall be evident when tested in accordance with method 5751.1 of Specification CCC-T-191.

3.1.1.1.2 *Water repellency.* The cotton cloth shall have a minimum spray rating of 50.0 when tested in accordance with method 5526 of Specification CCC-T-191.

3.1.2 *Components.* All components of the gas filter shall conform to the specifications and drawings listed on Drawing DL 5-19-673 and subsidiary drawings.

3.2 **Assembly.** The gas filter shall be assembled as specified in Drawing D5-19-673. During the assembly operation, the moisture content of the charcoal shall not exceed 4.0 percent when tested as specified in 4.6.1.

3.3 **Airflow resistance.** The airflow resistance shall be 4.5 — 0.5 inches of water at an airflow rate of 150 cubic feet per minute (cfm) when tested as specified in 4.6.2.

3.4 **Airflow resistance and phosgene gas life after rough handling.** After rough handling the filter for 15 minutes in a vertical position, as specified in 4.6.2, the airflow resistance shall be 4.5 ± 0.5 inches of water and the phosgene gas life shall be no less than 30 minutes at an airflow rate of 150 cfm and a phosgene concentration of 10 ± 2 milligram per liter of air when tested as specified in 4.6.3.

3.5 **Workmanship.** The gas filter shall be free from chipped, burred or bent metal sections or ripped or torn cloth or contamination by foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 *Government's responsibility.* The Government will be responsible for the performance of the tests in 4.6.3.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality assurance by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 have been satisfied.

4.2.3 *Substations inspection.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of Standard MIL-STD-105 are applicable to each substation and to the end item.

4.3 **Lotting.** A lot shall consist of the gas filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one sublot of charcoal sorbent shall be represented in any one lot of gas filters.

4.4 **Sampling.**

4.4.1 *For examination and nondestructive tests.* Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.4.2 *For destructive tests.* Sampling shall be conducted in accordance with Standard MIL-STD-414, Form 1, level IV, AQL 1.0 percent defective.

4.4.3 *For moisture content of charcoal sorbent.*

4.4.3.1 *Control container.* The control container shall have a minimum volume of 600 milliliters (ml). The inside height shall be approximately 2⅜ inches.

4.4.3.2 *Control container filling.* The control container shall be filled with sorbent from the filling apparatus just prior to the filling of the first ten, and each ten filters assembled thereafter each day. The uncovered control container shall precede the represented filters through the assembly line to the packaging station.

4.5 **Examination.**

4.5.1 *Examination and nondestructive tests.* Sample gas filters and the level A preparation of delivery shall be examined in accordance with the classification of defects and Standard MIL-STD-105.

4.5.2 *Critical defects.* Each gas filter shall be inspected for critical defects.

4.5.3 *Classification of defects.*

4.5.3.1 *Filter, gas, 150 CFM, M10 (Dwg D7-10-C73).*

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Categories	*Defects*	*Acceptable standards*
Critical		
1	Air resistance	4.6.2
Major	AQL 1.5 percent defective	
101	Component missing or incorrectly assembled	
102	Rubber seal on faces improperly attached	
103	Damage (see 3.5)	
104	Marking incorrect	
Minor	AQL 2.5 percent defective	
201	Sealing compound bolt ends missing	
202	Frame coatings missing inadequate or incorrect	
203	Foreign matter (see 3.5)	
204	Marking missing or illegible	

4.5.3.2 *Preparation for delivery.*

Categories	*Defects*
Critical	None defined
Major	AQL 2.5 percent defects
101	Containers incorrect
102	Filler pads missing or insufficient to make tight pads.
103	Packaging method incorrect
104	Quantity per container incorrect
105	Closure of container incorrect
106	Damaged container
107	Strapping loose or missing
108	Marking incorrect, missing, or illegible

4.6 Tests. Tests shall be conducted as follows:

4.6.1 *Moisture content of charcoal sorbent.* Determine the moisture content on approximately 25 gram specimen by drying for 15 minutes at 150° ± 7°C. using a number 276 Moisture Teller (see 6.4).

4.6.1.1 *Alternative moisture content method.* Determine the moisture content on 10 to 20 gram specimen weighed to the nearest milligram by drying in a tared flat bottom weighing dish, approximately 3 inches in diameter, for 3 hours in an oven at 150° ± 5°C. Cool in a desiccator and reweigh. Calculate the percent moisture as follows.

$$\text{Percent moisture} = \frac{100\ (A-B)}{W}$$

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when:

A = weight of dish and specimen before drying,

B = weight of dish and specimen after drying, and

W = weight of specimen.

4.6.2 *Resistance to airflow.* Determine the resistance to airflow using Penetrometer, Phosgene Sorption and Air Resistance of Gas Filters, Q108.

4.6.3 *Resistance to airflow and phosgene gas life after rough handling.* Rough handle the gas filter for 15 minutes in a vertical position using Machine, Vibrating, Filter Unit, Q110. Determine the airflow resistance and phosgene gas life using Penetrometer, Phosgene Sorption and Air Resistance of Gas Filters, Q108.

4.7 Acceptance/rejection criteria.

4.7.1 *Moisture content of charcoal.* If the moisture content of the charcoal does not meet the requirements of this specification, the 10 gas filters represented by the sample shall be rejected.

4.7.2 *Destructive tests.* Disposition of the lot represented by the samples for destructive test shall be determined in accordance with Standard MIL-STD-414, Form 1 and an AQL of 1.0 percent defective.

5. PREPARATION FOR DELIVERY

(For repair parts packaging see 6.6).

5.1 Packaging.

5.1.1 *Level A.* Two filters separated by a filler pad fabricated from the same material as the container cited herein, shall be packaged method II e (container barrier) of Specification MIL-P-116, except that a desiccant shall not be used. A class 1, style RSC fiber box, conforming to Specification PPP-B-636, with approximate inside dimensions of 24 3/16 by 24 3/16 by 5⅞ inches shall be used as the container. The packaged filters shall then be overwrapped with type E-1 or E-2 waterproof barrier-material, conforming to Specification UU-P-271, sealing all folds, laps and seams of the wrap with water-resistant adhesive conforming to Specification MIL-A-140. All sealed seams of the wrap shall be a minimum 2-inch width.

5.1.2 *Levels B and C.* Not applicable.

5.2 Packing.

5.2.1 *Level A.* Twelve filters packaged in units of two each, as specified in 5.1.1 shall be packed in a class 2, style 2, ½, 4 or 4½ nailed wood box conforming to Specification PPP-B-621. Closure shall be in accordance with the requirement of the box specifications, with strapping to be in accordance with the appendix thereto, using galvanized strapping.

5.2.2 *Levels B and C.* Not applicable.

5.3 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129. Marking shall include lot number and sequential number.

6. NOTES

6.1 Intended use. The gas filter covered by this specification is primarily intended for use with the ABC-M6A1 filter unit.

6.2 Ordering data. Procurement documents should specify the title, number and date of this specification.

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6.3 **Filling procedures.** Filters filled in the following manner have been found to satisfy the performance requirements of this specification.

(a) Place a filter frame with the bottom perforated sheet, fines filter, and draw-bolt assembly in a position for filling in a suitable gravity type filling machine.

(b) Feed a measured amount of properly screened sorbent gradually and uniformly from a three-foot height onto the filter frame by means of the filling machine. The weight of sorbent required to properly fill the frame will vary according to its bulk density and moisture content. The required weight has been found to be approximately 32 pounds per frame. It is advisable to predetermine experimentally the required amount of sorbent to fill a frame. (A volumetric measure will vary considerably less than a weigh measure). This is done by visual inspection of the mound made as described in paragraph (c) below and proved satisfactory by the phosgene life of a pair of filter, one rough handled and one not rough handled, tested in accordance with paragraph 4.6.2.

(c) After filling, place the frame on a table and carefully brush the surface sorbent back from the edge on each of the four sides to about two inches from the outer edges so as to form a bed level 18 inches square, elevated ¼ to ⅜ inch above the aluminum frame.

(d) Position the fines filter, perforated sheet, cup washer gasket, and cup washer on the draw bolt with the edges of the fines filter and perforated sheet properly aligned with the filter frame. Assemble the washer and nut to the draw-bolt, tightening the nut to an approximate torque of 40 inch-pounds. Clamp the edges of the perforated sheet to the aluminum frame with a suitable mechanical press and fasten the perforated sheets in place with thread cutting tapping screws.

(e) Remove the assembly from the press and tighten the draw bolt to an approximate torque of 85 inch-pounds. The draw bolt may require a greater or lesser torque. The nut should be tightened to the highest torque that will not cause splitting of the perforated sheet at the time of tightening or when the filter is rough handled.

(f) Complete the assembly as shown on Drawing D5-19-673.

6.4 **Test apparatus.** The No. 276 Moisture Teller is available from the Harry W. Dietert Co., Detroit, Michigan.

6.5 **International interest.** Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform the respective Departmental Standardization Officer so that appropriate action may be taken respecting the international agreement concerned.

6.6 **Repair parts packaging.** When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheet covering the gas filter should be used. (Copies of repair parts packaging data sheets may be obtained from the Di-

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rector for Engineering and Industrial Services, ATTN: Chief, Design Branch, U. S. Army Edgewood Arsenal, Edgewood Arsenal, Maryland.)

Custodians:

Army — MU

Navy — Marine Corps

International interest (see 6.5)

Preparing activity:

Army — MU

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